Control Charts for Time Series: A Review

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Abstract. A state of the art survey in Statistical Process Control (SPC) for dependent data will be given. First papers about the influence on and the modification of standard SPC schemes are written some decades ago. After that, in the end of the 1980s and during the 1990s the consideration of dependence in the field of SPC became very popular. It turned out, that falsely assuming independence mostly leads to improper SPC schemes. Therefore, one has to be aware of dependence.

Roughly speaking we can distinguish two different types of SPC schemes for dependent data, residual and modified schemes. The first ones are based on the standard schemes that are applied to the (empirical) residuals of a suitable time series model. For getting the latter ones the stopping rule of the scheme is modified. Furthermore, some special issues of CUSUM schemes in case of dependent data are considered.

In case of the most simple model of dependent data – the autoregressive process of order one – we compare modified and residual EWMA and CUSUM schemes.

We present some approximation methods for computation of SPC schemes characteristics. Furthermore, some theoretical relations between the schemes for dependent data and for independent data will be sketched.

1 Introduction

The objective of process monitoring, i. e. of Statistical Process Control (SPC), is to detect changes in a process that may result from uncontrollable and unpredictable causes at unknown times. The most important tools for monitoring data are control charts. In principle, a control chart is a decision rule which signals that the process has changed if the control statistic lies outside the control limits. The first control charts were introduced by Shewhart (1926, 1931). Since this pioneering work many other control schemes have been introduced, such as the cumulative sum (CUSUM) chart of Page (1954) and the exponentially weighted moving average (EWMA) chart of Roberts (1959).

Up to the end of the 1980s one of the basic assumptions of nearly all papers was that the underlying observations are independent over time. Only some papers dealt with the dependence problem. Goldsmith and Whitfield (1961) investigated the influence of correlation on the CUSUM chart by Monte Carlo simulations. In the same year Page (1961) mentioned: *"The effects of correlated observations have been examined in one case by Goldsmith and Whitfield; wider study is desirable."* The problem was addressed but one had to

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wait over 10 years more for the next papers. Then, Bagshaw and Johnson (1974,75,77) approximated the CUSUM statistic for correlated data by Brownian motions. Rowlands (1976) studied the performance of CUSUM schemes for AR(1) and ARMA(1,1) data (cf. Rowlands and Wetherill (1991)). Furthermore, Nikiforov (1975,79,83,84) wrote a lot about CUSUM charts for ARIMA data. Finally, Vasilopoulos and Stamboulis (1978) proposed modified alarm limits for the Shewhart chart in case of dependent data.

After the more theoretical oriented period of dealing with dependence of the data, in recent years it has been pointed out that quite often the independence assumption is incorrect. Wetherill (1977) reminded in his monography, that observations from modern industrial processes are often autocorrelated. Then, in the end of the 1980s treating the dependence problem became very popular. Alwan (1989) analyzed 235 datasets which were frequently used as (standard) SPC examples. It turned out that in about 85 % the control limits were misplaced. In nearly the half of these cases neglected correlation was responsible. Now, the problem was realized as severe.

Besides assessing these strong effects of correlation on the control charts as quality control tools in industrial statistics, one has to consider correlation as a typical pattern of data which we can find in medicine, environmental and financial statistics, and in automatic control in engineering. Here time series models are standard.

In the following, we present modifications of SPC schemes such as Shewhart chart, CUSUM procedure, and EWMA scheme in order to take into account the dependence.

First, we introduce the utilized time series models in Section 2. Then, it follows in Section 3 a short description of the (standard) control charts in the independent case. Moreover, we demonstrate the performance of the control charts designed for independent data and applied to AR(1) data.

In Section 4 and 5 we introduce the residuals and modified charts, respectively. There are two different types of SPC schemes for dependent data, residual and modified schemes. The first ones (e.g., Alwan and Roberts (1988), Wardell et al. (1994a,b)) are based on the standard schemes that are applied to the (empirical) residuals of a suitable time series model. It seems so, that Harrison and Davies (1964) were the first who applied control charts (CUSUM) to residuals (prediction errors). For getting the modified charts (e.g., Vasilopoulos and Stamboulis (1978), Schmid (1995, 1997a,b)) the stopping rule (variance term, control limit) of the scheme is modified. Furthermore, some special issues of CUSUM schemes (*keyword* likelihood ratio approach, e.g., Nikiforov (1975 and later), Schmid (1997b)) in case of dependent data are considered.

In Section 6 a comparison study is performed for modified and residual EWMA and CUSUM schemes for AR(1) data.

We restrict ourselves on the above charts because they are the most common charts for applications. Moreover, up to now it is difficult to find results for other SPC schemes in case of dependent data. In the following we use the average run length (ARL) as performance measure. The ARL denotes the average number of observations (or samples) from the beginning of the observation until the scheme signals. Thereby, it is assumed that the parameters are constant over time, i. e. a possible change takes place before the first observation was taken. In case of an undisturbed process the ARL should be large, otherwise small. Naturally, there are other measures too, but then the problem becomes much more complicate in presence of dependence.

2 Modeling

The aim of statistical process control (SPC) consists in detecting structural deviations in a process over time. It is examined whether the observations can be considered as realizations of a given target process $\{Y_t\}$. The procedure is a sequential one, i. e. a change should be detected immediately after its occurrence. The observed or actual process is denoted by $\{X_t\}$. In this paper it is always assumed that at each time point t exactly one observation is available. Thus the sample size is equal to 1. The extension to a larger sample size is straightforward.

For a long time it was always assumed that the random variables $\{Y_t\}$ are independent and identically distributed (iid). It has turned out that in applications this condition is frequently not fulfilled. The data have a memory. Within time series analysis many suitable models have been introduced to describe such a behavior. A good introduction to this topic is given, e. g., by Brockwell and Davis (1991) and Shumway and Stoffer (2000). In the next Subsection we present some basic results of this area which will be of importance in the following.

To assess the performance of a control procedure it is necessary to know the relationship between the target process $\{Y_t\}$ and the observed process $\{X_t\}$. Such (change-point) models are considered in the second part of this Section.

2.1 ARMA Processes

A time series is defined to be a collection of observations x_t, $t \in T_0$ indexed according to the order they are obtained in time. T_0 is equal to the set of time points of the observations. The observations are considered to be realizations of a stochastic process $\{X_t : t \in T\}$ with $T_0 \subset T$. In the following it is mainly assumed that $T = \mathbb{Z}$. As usually we use the term time series for both the process and a particular realization. No notational distinction is made between the two concepts.

A very important class of time series analysis are stationary processes. A stochastic process $\{X_t\}$ is called to be (weakly) stationary if

$$E(X_t) = \mu \quad \forall\, t \in \mathbb{Z}, \qquad Cov(X_t, X_{t+h}) = \gamma_h \quad \forall\, t, t+h \in \mathbb{Z}\ .$$

For stationary processes the mean function is constant over time and the covariance function between two time points depends only on their time spacing. Important characteristics of a stationary process in the time domain are the mean μ, the autocovariance function γ_h, and the autocorrelation function $\rho_h = \gamma_h/\gamma_0$.

The most popular family of time-correlated models are autoregressive moving average processes (ARMA). A stochastic process $\{X_t\}$ is called an ARMA process of order (p, q) with mean μ if it is a solution of the stochastic difference equation

$$X_t = \mu + \sum_{i=1}^{p} \alpha_i \,(X_{t-i} - \mu) + \varepsilon_t + \sum_{j=1}^{q} \beta_j \,\varepsilon_{t-j} \,. \tag{1}$$

Here it is assumed that $\{\varepsilon_t\}$ is a white noise process, i. e. that

$$E(\varepsilon_t) = 0, \quad Var(\varepsilon_t) = \sigma^2, \quad Cov(\varepsilon_t, \varepsilon_s) = 0 \quad \forall t \neq s \,.$$

In the case $p = 0$ the process $\{X_t\}$ is said to be a moving average process (MA(q)). For $q = 0$ an autoregressive process is given (AR(p)).

Now, of course, not all solutions of the equation (1) are stationary. It is possible to derive conditions on the coefficients such that there exists a stationary solution of the stochastic difference equation. This holds if all roots of the polynomial $1 - \sum_{i=1}^{p} \alpha_i \, z^i$ lie outside the unit circle (Brockwell and Davis (1991, p. 85)), i. e.

$$1 - \sum_{i=1}^{p} \alpha_i \, z^i \neq 0 \quad \forall z \in \mathbb{C} \,, \, |z| \leq 1 \,. \tag{2}$$

For an AR(1) process this condition reduces to $|\alpha_1| < 1$. For an AR(2) process the condition (2) is fulfilled if the coefficients α_1 and α_2 lie within the triangle which is built by $\alpha_1 + \alpha_2 < 1$, $\alpha_2 - \alpha_1 < 1$, and $|\alpha_2| < 1$. In the following we always assume that (2) holds.

If the coefficients of the ARMA process satisfy the condition (2) then there is exactly one stationary solution of the stochastic difference equation and it is given by

$$X_t = \mu \; + \; \sum_{i=0}^{\infty} s_i \, \varepsilon_{t-i} \,. \tag{3}$$

The coefficients $\{s_i\}$ can be determined by expanding the polynomial fraction $\left(1 + \sum_{j=1}^{q} \beta_j \, z^j\right) / \left(1 - \sum_{i=1}^{p} \alpha_i \, z^i\right)$ in a power series about the point $z = 0$. Then s_i is equal to the ith coefficient of the power series. If additionally $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ then $\{X_t\}$ is a Gaussian process. This means that the joint distribution function of every finite collection of the X_t's is a multivariate normal distribution.

Using the representation (3) of a stationary ARMA process its autocovariance function can be calculated (cf. Brockwell and Davis (1991, ch. 3)). For an AR(1) process it follows that

$$\gamma_0 = \frac{\sigma^2}{1-\alpha_1^2} \quad , \gamma_i = \alpha_1^{|i|}\,\gamma_0\,, \quad \rho_i = \alpha_1^{|i|}\,, \quad i \in \mathbb{Z}$$

and for an AR(2) process it holds that

$$\gamma_0 = \frac{\sigma^2\,(1-\alpha_2)}{(1+\alpha_2)\big((1-\alpha_2)^2-\alpha_1^2\big)}\,, \quad \gamma_1 = \frac{\sigma^2\,\alpha_1}{(1+\alpha_2)\big((1-\alpha_2)^2-\alpha_1^2\big)}\,,$$

$$\gamma_i = \alpha_1\,\gamma_{i-1} + \alpha_2\,\gamma_{i-2}\,, \quad i \geq 2\,.$$

Many papers about time series investigate the problem of forecasting future developments. It is desirable to have theoretically good predictors whose explicit calculation is simple. The theory of best linear predictors turns out to be simple, elegant and useful in practice (cf. Brockwell and Davis (1991, ch. 5)). If $\{X_t\}$ is a stationary process then the best (in the L^2 sense) linear predictor of X_t in terms of $X_1, \ldots, X_{t-1}$ is given by

$$\hat{X}_t = \mu + \sum_{i=1}^{t-1} \phi_{ti}\,(X_{t-i} \,-\, \mu)\,. \tag{4}$$

The coefficients $\phi_{t1}, \ldots, \phi_{tt-1}$ are real numbers which are obtained by solving the minimization problem

$$E\left(X_t - \mu - \sum_{i=1}^{t-1} b_{ti}\,(X_{t-i}-\mu)\right)^2 \quad \longrightarrow \quad \min_{b_{t1},\ldots,b_{tt-1}}\,.$$

It immediately follows that these weights fulfill the linear system of equations

$$\varGamma_t\,\boldsymbol{\phi}_t = \boldsymbol{\gamma}_t$$

where $\varGamma_t = (\gamma_{|i-j|})_{i,j=1,\ldots,t-1}$, $\boldsymbol{\gamma}_t = (\gamma_1, \ldots, \gamma_{t-1})'$ and $\boldsymbol{\phi}_t = (\phi_{t1}, \ldots, \phi_{tt-1})'$. The prediction error is given by $v_t = E\big(X_t - \hat{X}_t\big)^2 = \gamma_0 - \boldsymbol{\gamma}_t'\,\varGamma_t^{-1}\,\boldsymbol{\gamma}_t$ (Brockwell and Davis (1991, p. 168)). Recursive algorithms for determining the coefficients $\phi_{t1}, \ldots, \phi_{tt-1}$ can be found in Brockwell and Davis (1991, ch. 5).

For instance, it holds for an AR(p) process that for $t \geq p+1$

$$\hat{X}_t = \mu + \sum_{i=1}^{p} \alpha_i\,(X_{t-i} - \mu)\,.$$

The calculation of the predictors for the first p values shows to be more complicate but it always holds that $\hat{X}_1 = \mu$.

2.2 Modeling the Observed Process

By analyzing datasets we are frequently faced with structural changes of the underlying process like mean shifts, scale changes, outliers, drifts etc. Up to now in published literature about statistical process control for dependent data mostly changes in the mean and the variance of a time series have been investigated (there are some exceptions in literature on signal detection, see, e. g., the survey of Basseville (1988)). It is interesting that other types of changes like, e. g., changes in the correlation coefficients have been treated within econometrics (e. g., Johnston and DiNardo (1997)). However, in this context it is usually demanded that the time point of a possible change is known.

In this paper we restrict ourselves to deviations from the mean. A very general relationship between the target process $\{Y_t\}$ and the observed process $\{X_t\}$ is described by the model

$$X_t = Y_t + \sqrt{\gamma_0}\, f_t(\boldsymbol{a}, m)\, I_{\{m,m+1,\dots\}}(t)\,, \tag{5}$$

where I_A denotes the indicator function of the set A and $\gamma_0 = Var(Y_t)$. Consequently it holds that $X_t = Y_t$ for $t < m$. The influence of the change is described by the deterministic function $f_t(\boldsymbol{a}, m)$ which depends on m, the time index t, and the parameter vector $\boldsymbol{a}$. In the case $f_t(\boldsymbol{a}, m) = 0$ for all t it is said that the process $\{X_t\}$ is in control. The observed process and the target process coincide. Especially, it holds that $E(X_t) = E(Y_t)$ for all t. Sometimes $\{E(Y_t)\}$ is called the target sequence. If $E(Y_t) = \mu$ then μ is the target value. However, if $f_m(\boldsymbol{a}, m) \neq 0$, then there is a change at position m. The process $\{X_t\}$ is said to be out of control for $t \geq m$. In the case $f_t(\boldsymbol{a}, m) = a$ a shift in the mean is observed at position m. This model is the most widely discussed one in literature. If $\{Y_t\}$ is stationary then the process changes to another level for the mean. Setting $f_t(\boldsymbol{a}, m) = a\,(t - m + 1)$ a linear drift is present, while for $f_t(\boldsymbol{a}, m) = a_1\,(t - m + 1) + a_2\,(t - m + 1)^2$ a quadratic drift can be observed. Furthermore the choice $f_t(\boldsymbol{a}, m) = a\, I_{\{m\}}(t)$ leads to an outlier of type I (cf. Fox (1972)).

3 Standard Control Charts

The foundations of SPC were laid by Shewhart in the 1920s and 1930s. He introduced different control charts for monitoring the mean and the variance (e. g., Shewhart (1926,1931)). The Shewhart control charts do not have a memory. More precisely, the decision whether a change happened or not, is based only on the recent observation (sample). This is the main difference between Shewhart charts and the schemes of Page (1954) and Roberts (1959). Here, the influence of past data is controlled by an additional smoothing parameter.

For all these approaches it was always assumed that the data are independent. Therefore, we call these schemes standard control charts. Because these charts are well-known we only want to sketch them. For more details we confer to Montgomery (1991). Again, we restrict ourselves to the case with one single observation at each time point. We consider the following simplified change point model

$$X_t = Y_t + \sqrt{\gamma_0}\, a\, I_{\{m,m+1,\ldots\}}(t)\ , \quad t \geq 1\,. \tag{6}$$

The target value is given as $\mu = E(Y_t)$. We assume that μ and $\gamma_0 = Var(Y_t)$ are known. Usually, they are estimated by using a prerun (for $\{Y_t\}$). The process $\{X_t\}$ is in control if $a = 0$. Then $E(X_t) = E(Y_t) = \mu$ is valid for all $t \geq 1$. Otherwise, the process is out of control (for $t \geq m$).

3.1 EWMA Charts

The EWMA chart was introduced by Roberts (1959). It is based on the sequence

$$Z_t = (1-\lambda)\, Z_{t-1} + \lambda\, X_t\ , \quad t \geq 1\ . \tag{7}$$

The starting value Z_0 is set equal to z_0. Conventionally, the target value (of the mean) is chosen, i.e. $z_0 = E(Y_t) = \mu$. The parameter λ is named "smoothing parameter". It controls the influence of past data. For $\lambda = 1$ the EWMA chart becomes the Shewhart chart.

Solving (7) leads to

$$Z_t = z_0\,(1-\lambda)^t + \lambda \sum_{i=1}^{t} (1-\lambda)^{t-i}\, X_i\,.$$

This presentation shows that the smaller the smoothing parameter λ, the smaller the weight of the current observation and thus the larger the weight of the past data. The influence of the previous observations decreases exponentially (geometrically). For that reason the scheme is denoted as EWMA – exponentially weighted moving average, or in the past GMA – geometric moving average.

With $\sigma^2 = Var(Y_t)$ we obtain (cf. (6))

$$E(Z_t) = \mu + (1-\lambda)^t\,(z_0 - \mu) + a\,\sigma\left(1-(1-\lambda)^{t-m+1}\right) I_{\{m,m+1,\ldots\}}(t)\,,$$

$$Var(Z_t) = \frac{\lambda}{2-\lambda}\left(1-(1-\lambda)^{2t}\right)\sigma^2\,.$$

Thus,

$$\lim_{t\to\infty} E(Z_t) \underset{m<\infty}{=} \mu + a\,\sigma \quad , \quad \lim_{t\to\infty} Var(Z_t) = \frac{\lambda}{2-\lambda}\,\sigma^2\,.$$

In applications control charts based on the asymptotic variance prevailed. Their advantage is that the control limits are constant over time. One concludes that the process is out of control at time t if

$$Z_t > UCL = \mu + c\,\sigma\sqrt{\lambda/(2-\lambda)} \quad \text{or} \quad Z_t < LCL = \mu - c\,\sigma\sqrt{\lambda/(2-\lambda)}\,.$$

UCL and LCL denotes the upper and lower control limit, respectively. The critical value $c > 0$ is the second design parameter of the EWMA scheme (besides λ). The EWMA run length is given by

$$N_e = \inf\left\{t \in \mathbb{N} : Z_t > UCL \,\vee\, Z_t < LCL\right\}.$$

The most popular performance measure for control charts is the average run length (ARL). It is assumed that the process is out of control already at the beginning. This means that $m = 1$. Now, given a smoothing constant λ the critical value c is determined in order to get a prespecified in-control ARL, say ξ. Thus, one concludes on average after ξ observations that the process is out of control, although the process is in control. c is obtained as a solution of that implicit equation. In industrial statistics $\xi = 500$ is a common choice (or $\xi = 370$ as a result of the popular Shewhart $\bar{X}$ chart with 3σ limits), while in finance $\xi = 60$ might be an appropriate choice.

The computation of the ARL (as a function of the parameters λ and c) is quite complicate. Contrary to the Shewhart chart an explicit formula does not exist. Therefore, one has to use approximation techniques. The Markov chain approximation due to Brook and Evans (1972, for EWMA cf. Lucas/Saccucci (1990)) and the numerical solution of an integral equation (Crowder (1986)) are very popular techniques. Waldmann (1986a) presented a very accurate, but only seldom applied approximation method for computing the distribution of the run length of the EWMA scheme.

There remains the question about the choice of the parameter λ. Lucas and Saccucci (1990) determined for known a (out-of-control shift) the optimal λ. They minimized the out-of-control ARL. In their paper a lot of tables for independent and normally distributed data are given.

3.2 CUSUM Charts

CUSUM control charts were introduced by Page (1954). They are connected to the sequential probability ratio test (SPRT). Here, we derive CUSUM directly from the related SPRT.

Let $Y_t \sim \mathcal{N}(\mu, \sigma^2)$ for $t \geq 1$. Because of (6) it follows for $m = 1$ that $X_t \sim \mathcal{N}(\mu + a\,\sigma, \sigma^2)$. We denote with f_a the density of X_t. First, we consider the simple testing problem $H_0 : a = 0$ against $H_1 : a = a_0$ with known $a_0 > 0$. The SPRT says that sampling is stopped at time t if

$$\sum_{i=1}^{t} T_i \notin [A, B]$$

$$\text{with} \quad T_i = a_0\,\frac{X_i - (\mu + a_0\,\sigma/2)}{\sigma}\,.$$

If $\sum T_i > B$ then H_0 is rejected. Otherwise, if $\sum T_i < A$ then H_0 is accepted. The (random) sample number is given by

$$N = \inf\left\{ t \in \mathbb{N} : \sum_{i=1}^{t} T_i \notin [A, B] \right\} .$$

Based on the SPRT we get the standard CUSUM chart by setting $A = 0$, $h = B/a_0$, and $k = a_0/2$. Thus,

$$N = \inf\left\{ t \in \mathbb{N} : \sum_{i=1}^{t} (X_i - \mu - k\,\sigma) \notin [0, h\,\sigma] \right\} .$$

Now the CUSUM scheme can be obtained as follows. The SPRT is continued as long as H_0 is not rejected. The first rejection of the null hypothesis H_0 leads to a signal of the CUSUM chart. The first signal of the SPRT is given at

$$N_1 = \inf\left\{ t \in \mathbb{N} : \sum_{i=1}^{t} (X_t - \mu - k\,\sigma) \notin [0, h\,\sigma] \right\} .$$

If $\sum_{i=1}^{N_1}(X_i - \mu - k\,\sigma) > h\,\sigma$ then the SPRT rejects H_0. The CUSUM decision rule signals that the mean has changed from μ to $\mu + a\,\sigma$. Otherwise, if the first SPRT ends with $\sum_{i=1}^{N_1}(X_i - \mu - k\,\sigma) < 0$, then there is no reason to signal a parameter change and a new SPRT is started at time point $N_1 + 1$. Thereafter, the second stopping of the SPRT is at

$$N_2 = \inf\left\{ t > N_1 : \sum_{i=N_1+1}^{t} (X_i - \mu - k\,\sigma) \notin [0, h\,\sigma] \right\} .$$

The procedure is continued until the SPRT rejects H_0 for the first time. In other words, we add consecutively $X_i - \mu - k\,\sigma$. It must be taken into account that the part of the sum which leads to the stop below 0 has to be deleted. Hence, it is sufficient to consider

$$\begin{aligned} S_0^+ &= 0\,, \\ S_t^+ &= \max\left\{ 0, S_{t-1}^+ + (X_t - \mu - k\,\sigma) \right\} \quad \text{for } t \geq 1\,. \end{aligned}$$

The run length of the CUSUM chart is given by

$$N_c^+ = \inf\left\{ t \in \mathbb{N} : S_t^+ > h\,\sigma \right\} .$$

In the same way we can derive a decision rule for $a_0 < 0$. It is based on

$$\begin{aligned} S_0^- &= 0\,, \\ S_t^- &= \min\left\{ 0, S_{t-1}^- + (X_t - \mu + k\,\sigma) \right\} \quad \text{for } t \geq 1\,. \end{aligned}$$

Let N_c^- denote the corresponding stopping time.

If the sign of the change is unknown then both one-sided CUSUM charts are combined. The two-sided CUSUM chart gives a signal at time t if $S_t^+ > h\,\sigma$ or $S_t^- < -h\,\sigma$. We denote the run length of the two-sided scheme by $N_c = \min\{N_c^+, N_c^-\}$.

Again, the critical value h is chosen such that the in-control ARL is equal to a given quantity ξ. The parameter k is called reference value. Usually, a_0, i. e. k, is not known. If one is interested in detecting large changes then a_0 and k should be taken large. Otherwise, small shifts are detected faster if a_0 is small, too. Several simulations showed that for detecting a shift of size $a_0\,\sigma$ the best choice seems to be $k = |a_0|/2$. This is not surprising because of the optimality of the related SPRT for testing μ against $\mu + a_0\,\sigma$. This means that we have a hint how to determine the parameter k for the CUSUM chart. This is an advantage of the CUSUM chart because for the EWMA chart a similar formula is not known (there are some suitable approximations, see Srivastava and Wu (1997)).

As for the EWMA chart we can compute the ARL as a function of the chart parameters by using approximation techniques. We mention the Markov chain approach due to Brook and Evans (1972) which was originally developed for the CUSUM chart, the numerical solution of the ARL integral equation (e. g., Goel and Wu (1971), Vance (1986)), and the Waldmann approach (1986b). Starting with Roberts (1966) a lot of comparison studies were performed (cf. Lucas and Saccucci (1990), Srivastava (1994)). It turned out that for large changes, i. e. large a, the Shewhart charts have the best performance. For smaller shifts control charts with a memory like, e. g., the EWMA and the CUSUM chart lead to a smaller out-of-control ARL. There are only slight differences between them. Besides the ARL criterion some other measures for evaluating control charts (or change point detection schemes) are used. Considering a worst case and a min-max criterion Moustakides (1986) and Ritov (1990), respectively, showed that CUSUM is optimal. For more details on the CUSUM chart we confer to Hawkins and Olwell (1998).

3.3 Performance in the Presence of Autocorrelation

Frequently, SPC data are dependent. Therefore the question arises whether the standard control charts can be applied without any modification. In this Subsection we want to show the influence of correlation on the performance of the Shewhart, EWMA, and CUSUM chart.

We assume that the target process $\{Y_t\}$ is a stationary AR(1) process with mean 0. The white noise $\{\varepsilon_t\}$ is assumed to be normally distributed. Hence, the target value is equal to 0 and the variance of the target process is $\gamma_0 = \sigma^2/(1-\alpha_1^2)$.

In the following we assume that γ_0 is known. Consequently, the SPC user knows the variance of the target process. This assumption is not unusual. E. g., by using a (in-control) prerun it is possible to estimate the process variance. A natural estimator is the sample variance. This estimator is consistent and, unfortunately, biased. In Kramer and Schmid (2000) the influence of estimating unknown process parameters (see also Lu and Reynolds (1999)) on the ARL of Shewhart charts is investigated. They compare different variance estimators. Especially, they give recommendations for the sample size of the prerun. Here, we assume that the SPC user correctly determined the variance. That is, within the scope of our study we are only interested in the effects of neglecting correlation on the ARL.

Let ξ, i. e. the in-control ARL, be equal to 500. The smoothing parameter λ of the EWMA scheme and the reference value k of the CUSUM chart are determined such that a change of size $\sqrt{\gamma_0}$ (one unit measured in standard deviations, i. e. $a = 1$) is detected as fast as possible. This leads to $\lambda = 0.1$ (cf. Lucas and Saccucci (1990)) and $k = 0.5$ (see above). Now, suppose that the analyst falsely assumes that the data are independent. Then he will take the critical values from standard tables for independent normally distributed data. This procedure leads to the following stopping times N_s, N_e and N_c (Shewhart, EWMA, CUSUM)

$$N_s = \inf\left\{ t \in \mathbb{N} : |X_t| \geq 3.0902\sqrt{\gamma_0} \right\},$$

$$N_e = \inf\left\{ t \in \mathbb{N} : |Z_t| \geq 2.8143\sqrt{\lambda/(2-\lambda)}\sqrt{\gamma_0} \right\}$$
$$\text{with} \quad Z_0 = 0\,,\ \lambda = 0.1\,,$$

$$N_c = \inf\left\{ t \in \mathbb{N} : \max\{-S_t^-, S_t^+\} \geq 5.0707\sqrt{\gamma_0} \right\}$$
$$\text{with} \quad S_0^- = S_0^+ = 0\,,\ k = 0.5\,.$$

Let $E_a(N)$ denote the ARL of the control chart with stopping time N if in (6) (with $\gamma_0 = \sigma^2/(1-\alpha_1^2)$) a shift of size $a\sqrt{\gamma_0}$ at position $m = 1$ is present. Here we want to focus on the cases $a = 0$ and $a = 1$. Thus, for independent variables the in-control ARL $E_0(N)$ is equal to $\xi = 500$. In the out-of-control case the ARL is given by

$$E_1(N_s) = 54.6\,,\ E_1(N_e) = 10.3\,,\ E_1(N_c) = 10.5\,.$$

In Figure 1 we see the influence of the correlation on the ARL of the Shewhart chart. Remember, that $\alpha_1 = Corr(Y_t, Y_{t-1})$. The above graph shows the in-control ARL as a function of α_1. The in-control ARL is symmetric in α_1 around 0. Further, it is relatively robust for $|\alpha_1| \leq 0.5$. E. g., for $\alpha_1 = 0.4$ the relative deviation from the target ARL $\xi = 500$ is only 3.08 % and for $\alpha_1 = 0.5$ it is equal to 6.24 %. If the absolute value of α_1 is large then strong deviations from ξ can be noticed. A completely different behavior can be observed for the EWMA and the CUSUM chart (see Figure 2). Both schemes react very sensitively even to small correlations. For negative values of α_1 the in-control ARL is larger than assumed, for positive correlation the

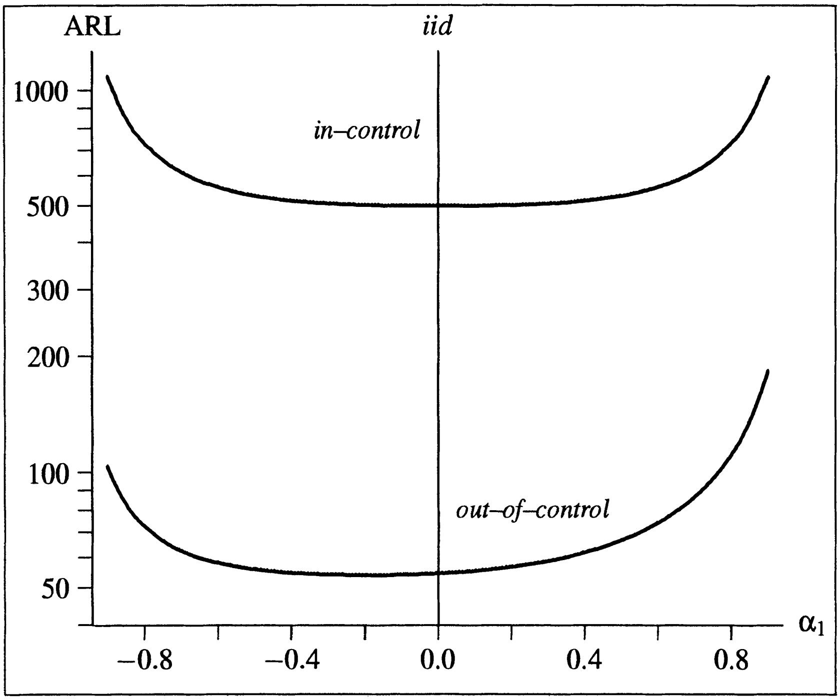

Fig. 1. ARL of the standard Shewhart chart in the presence of autocorrelation (model (6) with $m = 1$; $a = 0$: upper graph, $a = 1$: lower graph)

ARL is smaller. Already for $\alpha_1 = -0.1$ the relative deviation of $\xi = 500$ is equal to 100.6 % for the EWMA chart and even 123.9 % for the CUSUM chart. Although for positive correlations the deviations are not so large as for negative ones, nevertheless they can be dramatical (43.4 % and 47.6 %). Moreover, it can be stated that the CUSUM chart is more sensitive than the EWMA chart.

In the out-of-control situation the Shewhart chart is less robust with respect to correlation than in the in-control case. For $\alpha_1 = 0.5$ the deviation from the ARL of independent variables (i. e., $\alpha_1 = 0$) is equal to 21.9 %. The deviations are larger for positive than for negative autocorrelations. Again, the EWMA and the CUSUM chart behave different. It is interesting that they are more robust in the out-of-control state than in the in-control state. For $\alpha_1 = 0.5$ the deviation is merely 2.0 % (10.12 instead of 10.33) and 4.3 % (10.07 instead of 10.52), respectively. Note, that the graphs of the EWMA and the CUSUM schemes for $a = 1$ nearly coincide.

At first sight it is surprising that the Shewhart chart is more robust than the EWMA chart. In both cases the control statistic is compared with a target value. It is examined whether the deviation is larger than a multiple of the standard deviation of the statistic. Remember, that we assumed that

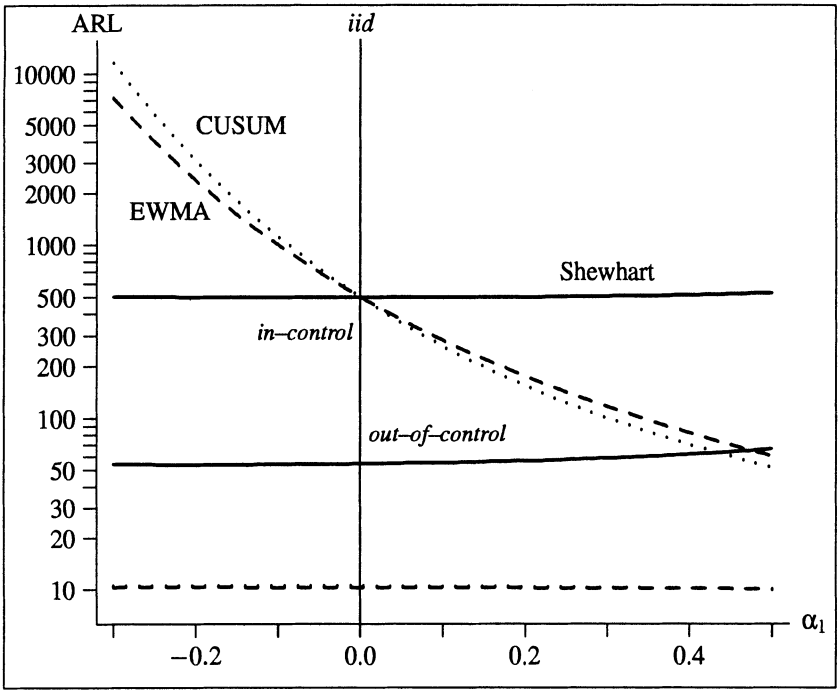

Fig. 2. ARL of the standard Shewhart (solid line), EWMA (dashed line) and CUSUM (dotted line) chart in the presence of autocorrelation (model (6) with $m = 1$; $a = 0$: upper graphs, $a = 1$: lower graphs)

the standard deviation of the target process is known. Consequently, we use the correct standard deviation for the Shewhart chart while the true variance of the EWMA statistic is not regarded (cf. Section 5.2). For that reason the EWMA chart may fail.

4 Residual Charts

Residual charts are a special type of control charts for time series. They have been intensively discussed in the last years (e. g., Alwan and Roberts (1988), Harris and Ross (1991), Montgomery and Mastrangelo (1991), Wardell et al. (1994a,b), Lu and Reynolds (1999)). Residual charts are based on the idea to filter the original data. The transformation is chosen such that the transformed quantities (residuals) are independent and identically distributed. Then all of the classical control schemes can be applied to the residuals. Suppose that (6) holds. The residuals $\hat{\varepsilon}_t = X_t - \hat{X}_t$, $t \geq 1$ are based on the comparison of the observation with a predictor $\hat{X}_t$ of X_t. Usually the best linear predictor of X_t (see Subsection 2.1) is taken. This quantity is calculated under the assumption that no shift in the mean is present ($a = 0$ in (6)). If

$\{X_t\}$ is stationary then $\hat{X}_t = \mu + \sum_{i=1}^{t-1} \phi_{ti} (X_{t-i} - \mu)$ (cf. (4)). Consequently it holds for $t \geq 1$ that

$$\hat{\varepsilon}_t = X_t - \mu - \sum_{i=1}^{t-1} \phi_{ti} (X_{t-i} - \mu).$$

If $\{Y_t\}$ is a Gaussian process then the random variables $\{\hat{\varepsilon}_t\}$ are independent in the in-control state, but they are not identically distributed. For that reason the residuals are normalized by the standard deviation $\sqrt{v_t}$ (cf. Subsection 2.1). The resulting variables $\hat{\Delta}_t = \hat{\varepsilon}_t / \sqrt{v_t}$, $t \geq 1$ are independent and identically distributed. It has to be emphasized that in the out-of-control state at least the second property does not hold any longer.

For an AR(1) process with mean μ it follows that $\hat{X}_t = \mu + \alpha_1 (X_{t-1} - \mu)$ for $t \geq 2$ and $\hat{X}_1 = \mu$. Consequently it holds that $\hat{\varepsilon}_t = X_t - \mu - \alpha_1 (X_{t-1} - \mu)$ for $t \geq 2$ and $\hat{\varepsilon}_1 = X_1 - \mu$. Because $Var(\hat{\varepsilon}_1) = \sigma^2/(1 - \alpha_1^2) \geq \sigma^2 = Var(\hat{\varepsilon}_t)$ for $t \geq 2$ the residuals have to be normalized. We get

$$\hat{\Delta}_t = \big(X_t - \mu - \alpha_1 (X_{t-1} - \mu)\big)/\sigma \quad \text{for } t \geq 2 \text{ and}$$
$$\hat{\Delta}_1 = \sqrt{1 - \alpha_1^2}\,(X_1 - \mu)/\sigma.$$

Now in the in-control state the variables $\hat{\Delta}_t$, $t \geq 1$ are independent and identically distributed. All control charts of Section 3 can be directly applied to the normalized residuals $\{\hat{\Delta}_t\}$. If the process is out of control from the beginning then $E(X_t) = \mu + a\sqrt{\gamma_0}$ for $t \geq 1$ and

$$E(\hat{\Delta}_t) = a\sqrt{\frac{1-\alpha_1}{1+\alpha_1}} \quad \text{for} \quad t \geq 2, \qquad E(\hat{\Delta}_1) = a. \tag{8}$$

If $\alpha_1 \geq 0$ then $E(\hat{\Delta}_t) \leq a$ and for $\alpha_1 \leq 0$ we get $E(\hat{\Delta}_t) \geq a$. Consequently the effect of the transformation is that for positive coefficients the influence of the change $a\sqrt{\gamma_0}$ decreases while it increases for negative coefficients.

In many papers the starting problem is not treated. But this procedure is not satisfactory because then the knowledge of values X_t, $t \leq 0$ is tacitly demanded. Taking $\hat{\varepsilon}_1 = X_1 - \alpha_1 X_0$ it is assumed that X_0 is known. With respect to the ARL this means a conditional ARL is determined. A thorough discussion of the starting problem is given by Kramer and Schmid (1997).

As an extension of the residual charts the so-called ARMA charts (Jiang et al. (2000)) can be considered. Here the general intention is to transform the data in order to detect a change faster.

5 Modified Control Charts

Residual charts are based on a transformation of the observations. This is a completely different attempt than for modified schemes. The idea behind this type of charts is to extend the control design of the classical charts directly to the case of correlated data. Thus, this method seems to be more devoted to practitioners.

5.1 Modified Shewhart Charts

Let $\{Y_t\}$ be a stationary process with mean μ and autocovariance function $\{\gamma_h\}$. The relationship between the target process and the observed process is again described by model (6) with $m = 1$. Then it is concluded that the process is out of control at time t if

$$|X_t - \mu| > c\sqrt{\gamma_0}\,.$$

In the following it is assumed that both μ and γ_0 are known. The decision rule is based on the comparison of the observation at time t with the target value μ. If the distance between both quantities is larger than a certain multiple of the standard deviation then the process is declared to be out of control. This is the same idea as for the standard Shewhart charts. The only difference is the use of the variance of the time series (and a possibly different critical value c). The run length of this chart is denoted by

$$N_s = \inf\left\{t \in \mathbb{N} : |X_t - \mu| > c\sqrt{\gamma_0}\right\}.$$

This control chart was proposed by Vasilopoulos and Stamboulis (1978). A thorough discussion is given in Schmid (1995).

The control statistics of the modified schemes are not independent. Therefore the analytical analysis of these charts is more difficult than in the case of independent samples. In the next theorem the run length of the control chart for correlated data is compared with that for independent observations. We use the symbol P_{iid} to indicate that the process $\{Y_t\}$ is assumed to be independent and identically distributed.

Theorem 1: Let $\{Y_t\}$ be a Gaussian process with mean μ and variance $\gamma_0 > 0$. Let $k \in \mathbb{N}$. Then it holds in the in-control state that

$$P(N_s > k) \;\geq\; P_{iid}(N_s > k) \;=\; \left(2\Phi(c) - 1\right)^k. \tag{9}$$

Proof: see Schmid (1995).

In Theorem 1 the event is considered that an in-control process is assessed to be in control at least up to time k. On the right hand side of (9) this probability is determined for iid variables with mean μ and variance γ_0. The theorem says that for correlated data the probability of the event is greater or equal than for independent ones. Consequently, if a practitioner falsely assumes that the underlying data are independent and identically distributed, then he will determine c according to $c_{iid} = \Phi^{-1}\left(1 - 1/(2\xi)\right)$. Theorem 1 says that the true value of c is less or equal to c_{iid}.

Theorem 2: Let $\{Y_t\}$ be a stationary AR(1) process. Assume that the variables $\{\varepsilon_t\}$ are independent and normally distributed with mean 0 and variance σ^2. Then it holds that in the in-control state $P(N_s > k)$ is a nondecreasing function in $|\alpha_1|$.

Proof: see Schmid (1995).

A discussion of the behavior in the out-of-control state is given by Schmid (1995). It is proved for a large family of distribution functions that given the model (5) the out-of-control ARL is smaller or equal to the in-control ARL.

Now $E(X_t - \mu)/\sqrt{\gamma_0} = a$. For an AR(1) process it holds that for a positive autocorrelation coefficient $E(\hat{\Delta}_t) \leq a$ for $t \geq 1$ (cf. (8)). This shows that the shift has a greater effect on the (modified) Shewhart chart than on the residual chart. This leads to the conjecture that the modified Shewhart chart will provide better results than the classical Shewhart scheme applied to the residuals.

Explicit formulas of the ARL for correlated data are at the moment only available for exchangeable variables (Schmid (1995)). For AR(1) processes an approximation of the ARL can be determined by the Markov chain approach of Brook and Evans (1972). For more complex processes, however, estimators for the ARL can only be obtained by simulations.

5.2 Modified EWMA Charts

The modified EWMA chart was introduced by Schmid (1997a). It is based on the recursion

$$Z_t = (1 - \lambda)\, Z_{t-1} + \lambda\, X_t$$

for $t \geq 1$. Let $Z_0 = z_0$ and $\lambda \in (0, 1]$. The target process $\{Y_t\}$ is assumed to be a stationary process with mean μ and autocovariance function $\{\gamma_v\}$.

Lemma 1: Let $\{Y_t\}$ be a stationary process with mean μ and autocovariance function $\{\gamma_v\}$. Furthermore suppose that (5) holds.

a) If $f_t(a, m) = a\, g_t(m)$ then it holds for $t \geq m$ that

$$E(Z_t) = \mu + (1 - \lambda)^t\, (z_0 - \mu) + \lambda \sum_{j=0}^{t-m} (1 - \lambda)^j\, g_{t-j}(m)\,.$$

b)

$$\sigma^2_{e,t} := Var(Z_t) = \lambda^2 \sum_{|i| \leq t-1} \gamma_i \sum_{j=\max\{0,-i\}}^{\min\{t-1, t-1-i\}} (1 - \lambda)^{2j+i}\,.$$

If $\{\gamma_v\}$ is absolutely summable then it holds that

$$\sigma^2_{e,t} \rightarrow \sigma^2_e := \frac{\lambda}{2 - \lambda} \sum_{j=-\infty}^{\infty} \gamma_j\, (1 - \lambda)^{|j|}$$

as $t \rightarrow \infty$.

c)

$$Cov(Z_t, Z_{t+h}) = \lambda^2 \sum_{i=-(t+h-1)}^{t-1} \gamma_i \sum_{\tau=\max\{0,-h-i\}}^{\min\{t-1-i,t-1\}} (1-\lambda)^{2\tau+i+h}$$

Proof: The results can be obtained by straightforward calculations.

For instance, it follows that for an AR(1) process

$$\sigma_e^2 = \sigma^2 \frac{\lambda}{2-\lambda} \frac{1+\alpha_1(1-\lambda)}{1-\alpha_1(1-\lambda)} .$$

Putting $\alpha_1 = 0$ the variance for independent variables is obtained. If $\{Y_t\}$ has a spectral density f then we get

$$\sigma_e^2 = \lambda^2 \int_{-\pi}^{\pi} \frac{f(x)}{1-2(1-\lambda)\cos x + (1-\lambda)^2}\, dx .$$

It is concluded that the process is out of control at time t if

$$|Z_t - \mu| > c\sigma_e .$$

This decision criterion is a direct extension of the design of the standard EWMA chart. To simplify the scheme the asymptotic variance is used again. If $\gamma_v = 0$ for $v \neq 0$ then the classical chart of Roberts is present. Choosing $\lambda = 1$ the chart reduces to the modified Shewhart chart.

A comparison of the run length of the EWMA chart for correlated and for independent variables is given by Schmid and Schöne (1997). They deal with a one-sided control chart whose run length is given by

$$N_e^u = \inf\left\{t \in \mathbb{N} : Z_t - \mu > c\sqrt{Var(Z_t)}\right\} .$$

Note that here the variance at time t is used and not the asymptotic variance. The next theorem can be considered to be a generalization of Theorem 1, however, for the one-sided problem.

Theorem 3: Let $\{Y_t\}$ be a stationary Gaussian process with mean μ and autocovariance function $\{\gamma_v\}$ with $\gamma_v \geq 0$ for all v. Let $z_0 = \mu$. Then it holds for $k \in \mathbb{Z}$ that

$$P(N_e^u > k) \geq P_{iid}(N_e^u > k) .$$

Proof: see Schmid and Schöne (1997).

In Theorem 3 the index *iid* means that the target process $\{Y_t\}$ is assumed to be independent and identically distributed with mean μ and variance γ_0. Note that this does not imply that the EWMA statistics $\{Z_t\}$ are independent. Even for independent variables $\{Y_t\}$ they are correlated (cf. Lemma 1, c)). Furthermore, Theorem 3 is no longer valid if the autocovariances may be negative (see Schmid and Schöne (1997)).

In Theorem 3 the ARL of the EWMA chart in the case of correlated observations is compared with its ARL for independent variables. This result was generalized in Schöne et al. (1999). They proved that the ARL shows a monotonicity behavior in the autocovariances.

Let $\{Y_t\}$ be a Gaussian process with mean μ and autocovariance function $\gamma = \{\gamma_v\}$. In order to express the dependence upon the autocovariances we use the notation P_γ, E_γ and Var_γ.

Theorem 4: Let $\{Y_t\}$ be a stationary Gaussian process with mean μ and autocovariance functions $\{\gamma_v\}$ and $\{\delta_v\}$, respectively. Assume that

$$\gamma_0 > 0,\ \delta_0 > 0,\ \gamma_h \geq 0,\ \delta_h \geq 0 \quad \text{for} \quad 1 \leq h \leq k-1\,,$$

and

$$\gamma_i\,\delta_j \quad \geq \quad \gamma_j\,\delta_i \quad \text{for} \quad 0 \leq j < i \leq k-1\,.$$

Then it holds for all real numbers $c_1, \ldots, c_k$ that

$$P_\gamma\left(Z_t - E_\gamma(Z_t) \leq c_t\sqrt{Var_\gamma(Z_t)}\ ,\ t = 1, \ldots, k\right)$$
$$\geq P_\delta\left(Z_t - E_\delta(Z_t) \leq c_t\sqrt{Var_\delta(Z_t)}\ ,\ t = 1, \ldots, k\right)\,.$$

Proof: see Schöne et al. (1999).

The design of the modified control charts is simpler than that of the residual charts. The main problem consists in the determination of the critical values c, where c is taken such that the in-control ARL is equal to a pre-specified value. As already mentioned in Section 3.1 the calculation of the critical values turns out to be difficult even for independent variables. If the data show a correlation structure then no explicit formula for the ARL is available. Simulations are necessary. The critical values depend not only on the given value for the in-control ARL ξ, but on all parameters of the target process, too. For that reason tables have to be determined for each process.

What recommendations can be made about the choice of the smoothing parameter λ? As in the case of independent variables there is, of course, no λ which provides the smallest out-of-control ARL for all shifts. However, again the rule of thumb holds that λ should be taken small if a small shift is expected and a large value of λ is better for detecting larger shifts. If the magnitude of the expected shift is known then λ can be chosen such that the corresponding out-of-control ARL takes its minimum. Such tables can be found in Schmid (1997a). An impression about the relationship between the optimal value of λ and the correlation can be obtained by the results in Section 6, too.

A generalization of the modified EWMA chart to the multivariate case is given in Kramer and Schmid (1997).

5.3 Modified CUSUM Charts

In the literature different approaches were used to extend CUSUM charts to time series data. Nikiforov (1983) applied the SPRT to ARMA processes with

Table 1. Critical values of the EWMA chart for an AR(1) process ($\xi = 500$) *(see Kramer (1997))*

λ	α_1								
	-0.8	-0.6	-0.4	-0.2	0.0	0.2	0.4	0.6	0.8
1	2.965	3.055	3.081	3.088	3.090	3.088	3.081	3.055	2.965
0.8	3.004	3.074	3.088	3.090	3.089	3.082	3.064	3.024	2.917
0.6	3.036	3.085	3.090	3.087	3.081	3.064	3.035	2.979	2.859
0.4	3.057	3.084	3.082	3.072	3.054	3.024	2.979	2.910	2.779
0.2	3.048	3.049	3.028	2.997	2.962	2.915	2.857	2.777	2.638
0.1	2.978	2.942	2.899	2.856	2.814	2.764	2.705	2.625	2.487
0.05	2.820	2.753	2.701	2.657	2.615	2.570	2.517	2.445	2.316
0.025	2.574	2.493	2.444	2.405	2.368	2.332	2.288	2.227	2.116
0.01	2.142	2.069	2.030	2.001	1.973	1.949	1.917	1.872	1.788

normal white noise. Yashchin (1993) compared CUSUM schemes based on the original time series data with suitable modified control limits and CUSUM schemes applied to transformed data (similar to the residual chart approach). For the latter transformation Yashchin exploited the same likelihood ratio as later Schmid (1997b). For computational reasons he simplified that ratio. In order to compute ARLs for the first schemes he used an approximation technique, where the original data were replaced by iid data with similar properties. Schmid (1997b) made use of Siegmund's (1985) way of introducing CUSUM. Now, the target process is assumed to be an arbitrary Gaussian process. For independent data all approaches lead to the same CUSUM chart. However, in case of autocorrelated data slightly different schemes are obtained. Note, that Sparks (2000) considered further CUSUM adaptions for AR(1) data, i. e. adjusted reference value k and a modified restarting mechanism.

In the following the approach of Schmid (1997b) is described. For all $t \in \mathbb{N}$ let $\boldsymbol{Y}_t = (Y_1, \ldots, Y_t)'$ be normally distributed with expectation $\boldsymbol{\mu}_t = \mu\,(1, \ldots, 1)'$ and (positive definite) correlation matrix C_t. Furthermore, we assume that $\gamma_0 = Var(Y_t)$ for all t (homoscedasticity). γ_0, $\boldsymbol{\mu}_t$ and C_t are assumed to be known. With f_0 and $f_{a,m}$ we denote the density of the observed vector $\boldsymbol{X}_t$ in case of no change point at all and a change point at position m, respectively. Then it follows that

$$-2\log\frac{f_0(\boldsymbol{X}_t)}{\max\limits_{1\le m\le t+1} f_{a,m}(\boldsymbol{X}_t)} = 2\max\left\{0,\ \max_{1\le m\le t} a\,\frac{T_{mt} - a\sqrt{\gamma_0}\,K_{mt}/2}{\sqrt{\gamma_0}}\right\}$$

with $T_{mt} = e_{mt}'\,C_t^{-1}\,(\boldsymbol{X}_t - \boldsymbol{\mu}_t)$ and $K_{mt} = e_{mt}'\,C_t^{-1}\,e_{mt}$. e_{mt} denotes the vector with zeros in the first $m-1$ rows and ones in the remaining $t-m+1$ rows.

The process stops at time t if

$$\max_{1\le m\le t}\left\{T_{mt}-k\sqrt{\gamma_0}\,K_{mt},-(T_{mt}+k\sqrt{\gamma_0}\,K_{mt})\right\}>h\sqrt{\gamma_0}\,.$$

As in the iid case the reference value k is taken equal to $|a|/2$.

The practical computation of this stopping rule is time consuming. However, it is possible to compute the statistic recursively (cf. Schmid (1997b)). E. g., for an AR(1) process with expectation μ we obtain the following run length

$$N_c=\inf\left\{t\in\mathbb{N}:S_t^+>h\sqrt{\gamma_0}\quad\text{or}\quad S_t^-<-h\sqrt{\gamma_0}\right\}\tag{10}$$

with

$$\begin{aligned}
S_t^+&=\max\left\{0,\varepsilon_t-k\sqrt{\gamma_0},S_{t-1}^++(1-\alpha_1)\left(\varepsilon_t-(1-\alpha_1)k\sqrt{\gamma_0}\right)\right\}\ ,\ t\ge 2\,,\\
S_1^+&=\max\left\{0,(1-\alpha_1^2)\,(X_1-\mu-k\sqrt{\gamma_0})\right\},
\end{aligned}$$

$$\begin{aligned}
S_t^-&=\min\left\{0,\varepsilon_t+k\sqrt{\gamma_0},S_{t-1}^-+(1-\alpha_1)\left(\varepsilon_t+(1-\alpha_1)k\sqrt{\gamma_0}\right)\right\}\ ,\ t\ge 2\,,\\
S_1^-&=\min\left\{0,(1-\alpha_1^2)\,(X_1-\mu+k\sqrt{\gamma_0})\right\}\ .
\end{aligned}$$

The SPRT approach (cf. Nikiforov (1975 and later)) is based on comparing $f_0(\boldsymbol{X}_t)$ with $f_{a,1}(\boldsymbol{X}_t)$. This leads to

$$T_t\ =\ \log\frac{f_{a,1}(\mathbf{X}_t)}{f_0(\mathbf{X}_t)}\ =\ a\,\frac{T_{1t}-a\sqrt{\gamma_0}\,K_{1t}/2}{\sqrt{\gamma_0}}\,.$$

The related CUSUM chart gives a signal at time t if

$$\max_{1\le m\le t}\left\{T_{1m}-k\sqrt{\gamma_0}\,K_{1m},-(T_{1m}+k\sqrt{\gamma_0}\,K_{1m})\right\}>h\sqrt{\gamma_0}\,.\tag{11}$$

For AR(1) data similar recursions as in (10) are available. The main difference is the vanishing of the middle term $\varepsilon_t\mp k\sqrt{\gamma_0}$ in the recursion rules for S_t^+ and S_t^-, respectively.

The reference value k is selected as in the iid case. For both CUSUM charts (10) and (11) the design parameter h can be determined by the Markov chain approximation (cf. again Brook and Evans (1972)) or for complex processes by simulations. Note that in the next Section (comparison study) the Markov chain approach is used.

Several tables of critical values for the modified EWMA chart and the modified CUSUM chart (10) due to Schmid can be found in Kramer (1997). These values were computed by Monte Carlo simulations for an in-control ARL of $\xi=500$. For a different approach of computing the ARL of CUSUM schemes in case of autoregressive processes we refer to Rowlands (1976). Further, we refer to a recent paper of Lu and Reynolds (2001), who considered modified and residual CUSUM and EWMA schemes for an ARMA(1,1) by computation of the steady-state ARLs.

6 Comparison Study

Comparisons of the behavior of control charts for time series were given by several authors (e. g., Kramer (1997), Schmid (1997b), Kramer and Schmid (1998), Lu and Reynolds (2001)). In this paper we want to compare the control charts described in Section 4 and Section 5. As a measure for the performance the ARL is used which is determined by using Monte Carlo simulations and Markov chain approximations, respectively.

The target process $\{Y_t\}$ is given by an autoregressive process of order 1. Let

$$Y_t = \alpha_1 Y_{t-1} + \varepsilon_t$$

with $\varepsilon_t \sim \mathcal{N}(0,1)$ and $Cov(\varepsilon_s, \varepsilon_t) = 0$ for $s \neq t$. Thus, the target value is equal to 0. The relation between the target process and the observed process is given by the change point model (6) with $m = 1$

$$X_t = Y_t + a\sqrt{\gamma_0}$$

where $a \in \{0, 0.5, 1.0, \ldots, 3.0\}$ and $Var(X_t) = \gamma_0 = 1/(1-\alpha_1^2)$. The following five control charts

(a) modified EWMA chart (cf. 5.2), abbr. EWMAmod,
(b) residual EWMA chart (cf. 4 and 3.1), abbr. EWMAres,
(c) modified CUSUM chart due to Schmid (cf. 5.3), abbr. CUSUMmod,
(d) residual CUSUM chart (cf. 4 and 3.2), abbr. CUSUMres,
(e) CUSUM chart with modified control limits, abbr. CUSUMcla

are adjusted in order to achieve an in-control ARL of $\xi = 500$. The remaining chart parameters (λ for EWMA, k for CUSUM) are taken such that the out-of-control ARL for a given shift a is minimal. Here for EWMAmod λ takes values within $\{0.01, 0.025, 0.05, 0.1, 0.2, 0.4, 0.6, 0.8, 1\}$ (cf. Table 1 or Kramer (1997)) and for EWMAres additionally the values $\{0.3, 0.5, 0.7, 0.9\}$ are permitted. The reference value k for the CUSUM charts is chosen from $\{0, 0.1, 0.2, \ldots, 2.0\}$ for CUSUMmod and CUSUMres, and more roughly from $\{0, 0.25, 0.5, \ldots, 2.0\}$ for CUSUMcla (cf. Kramer (1997)). The ARLs are computed by Monte Carlo simulations for the modified charts EWMAmod and CUSUMclass (Kramer (1997); 100,000 repetitions) and by the Markov chain approach (Brook and Evans (1972) or Lucas and Saccucci (1990)) with matrix dimension 300 for the other charts. Then, the critical values are obtained by iteration algorithms like the regula falsi. After that, the critical values were validated by a Monte Carlo simulation with 1,000,000 repetitions. The ARLs in the out-of-control case for EWMAmod and CUSUMmod are computed by Monte Carlo simulations with 10,000,000 repetitions while for the remaining charts the Markov chain approach was used again.

Now, for all constellations of a and of the control chart parameters λ and k, respectively, the out-of-control ARLs were computed and compared.

Table 2. Critical values of the (standard) CUSUM chart with modified control limits (CUSUMcla) for AR(1) data ($\xi = 500$) *(from Kramer (1997))*

k	α_1								
	-0.8	-0.6	-0.4	-0.2	0.0	0.2	0.4	0.6	0.8
0	32.896	38.994	39.096	35.831	30.481	24.122	17.246	10.494	4.576
0.25	3.846	6.497	8.078	8.737	8.575	7.878	6.588	4.814	2.671
0.5	1.909	3.459	4.489	5.014	5.066	4.826	4.204	3.223	1.904
0.75	1.119	2.258	3.017	3.437	3.538	3.455	3.078	2.414	1.430
1.0	0.854	1.520	2.205	2.565	2.664	2.657	2.405	1.896	1.075
1.25	0.604	1.226	1.638	1.995	2.105	2.130	1.941	1.519	0.808
1.5	0.356	0.972	1.342	1.581	1.707	1.742	1.594	1.231	0.637
1.75	0.111	0.723	1.083	1.293	1.389	1.421	1.315	1.016	0.539
2.0	.[1]	0.473	0.832	1.031	1.109	1.200	1.099	0.859	0.474

[1]: It was not possible to find a value h for $k = 2$ and $\xi = 500$. Therefore, as an approximation $h = 10^{-7}$ was used which led to an in-control ARL of 1120.

The minimal out-of-control ARL for each scheme is given in Table 3. For each possible shift a the corresponding parameter value is shown, too. For a fixed shift the ARL of the fastest chart is bold typed. Furthermore, for autocorrelations $\alpha_1 \in \{-0.6, 0, 0.6\}$ the minimal out-of-control ARLs were plotted in Figure 3.

At first sight, for positive α_1 the modified charts (EWMAmod, CUSUM-cla) dominate while for negative α_1 the residual charts provide smaller out-of-control ARLs. With some exceptions ($a = 0.5$ and $\alpha_1 = -0.8$, $a = 1$ and $\alpha_1 = -0.6$, $a = 1$ and $\alpha_1 = -0.4$) we observe the well-known fact that EWMA charts detect small changes somewhat faster than CUSUM charts (remember that we measure by the ARL). The CUSUM charts are faster for shifts a larger than 1 (exceptions: $a = 3$ and $\alpha_1 \in \{-0.4, -0.2, 0.8\}$, $a = 2.5$ and $\alpha_1 = -0.4$).

If we consider the optimal design parameters then it can be observed for the residual charts (EWMAres, CUSUMres) that the values are similar to the optimal design parameters of the standard charts for a shift of size $(1 - \alpha_1)\,a$. For CUSUMres the deviations are smaller than for EWMAres. The chart CUSUMmod which is related to a residual chart shows the most convenient behavior since the optimal k does not depend on α_1. Exceptions can only be observed for strong correlations ($|\alpha_1| = 0.8$). The optimal design parameters of the modified charts behave similarly than those of the residual charts, but are more stable with respect to α_1.

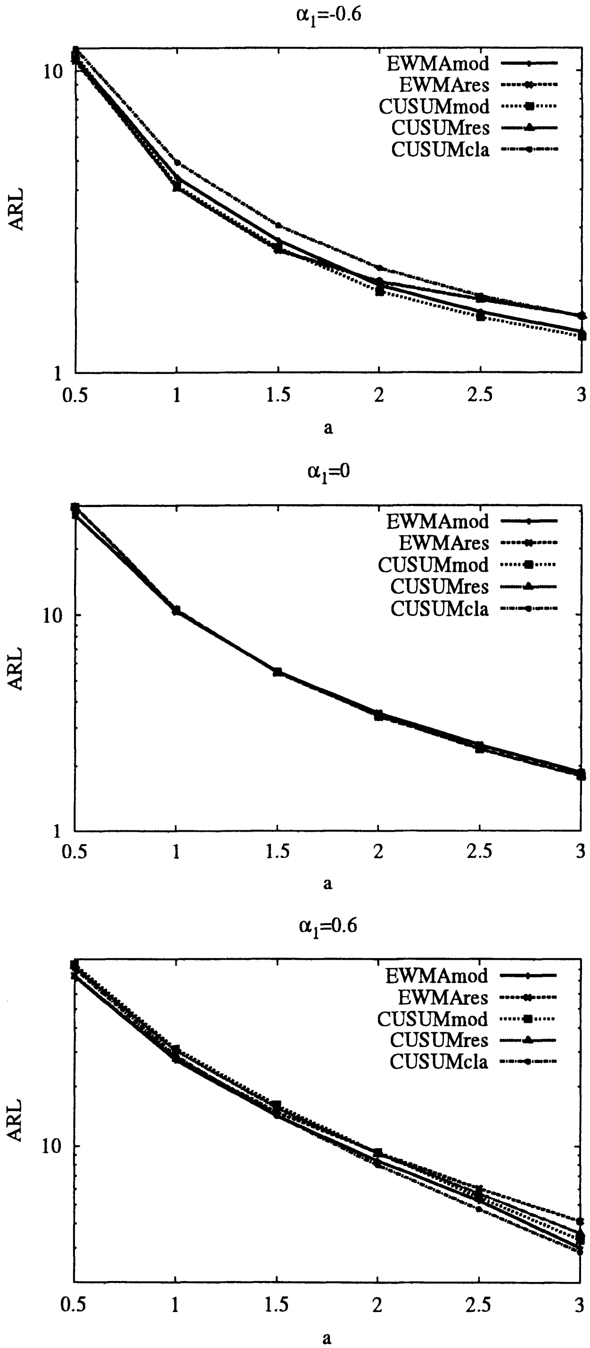

Fig. 3. ARLs in the out-of-control case of the 5 control charts under consideration for AR(1) data; lag-1-autocorrelation α_1 is equal to -0.6, 0, and 0.6.

Table 3. Average Run Length (ARL) in the out-of-control case of the 5 control charts under consideration for AR(1) data (model (6), $\xi = 500$, in parentheses: optimal values for λ and k, respectively)

α_1	chart	shift a 0.5	1.0	1.5	2.0	2.5	3.0
-0.8	EWMAmod	(.1)7.137	(.2)3.193	(.2)2.073	(.4)1.630	(.4)1.408	(.4)1.231
	EWMAres	(.2)6.166	(.6)**2.644**	(.8)2.004	(1)1.864	(1)1.722	(1)1.536
	CUSUMmod	(.3)6.381	(.5)2.681	(1.8)**1.737**	(1.8)**1.486**	(1.8)**1.295**	(1.8)**1.150**
	CUSUMres	(.4)**6.154**	(.8)2.668	(1.1)2.015	(1.8)1.864	(1.8)1.722	(1.8)1.536
	CUSUMcla	(.25)9.382	(.25)3.923	(.25)2.665	(.25)1.960	(.5)1.686	(.75)1.486
-0.6	EWMAmod	(.1)11.102	(.2)4.389	(.4)2.721	(.4)1.938	(.4)1.583	(.4,.6)1.363
	EWMAres	(.1)**10.823**	(.4)4.058	(.6)**2.514**	(.8)1.988	(.9)1.741	(1)1.537
	CUSUMmod	(.3)11.364	(.5)4.128	(.7)2.573	(2.4)**1.851**	(2.4)**1.521**	(2.4)**1.314**
	CUSUMres	(.4)10.964	(.8)**4.053**	(1.1)2.540	(1.4)1.997	(1.8)1.740	(2.4)1.537
	CUSUMcla	(.25)11.962	(.25)4.930	(.5)3.051	(.5)2.206	(.5)1.787	(.75)1.531
-0.4	EWMAmod	(.1)15.947	(.2)5.854	(.4)3.456	(.4)2.339	(.4)**1.824**	(.6)**1.482**
	EWMAres	(.1)**15.738**	(.2)5.716	(.4)3.255	(.6)2.319	(.8)1.863	(.9)1.567
	CUSUMmod	(.3)16.900	(.5)5.817	(.7)3.313	(1.2)2.353	(1.3)1.861	(2.4)1.500
	CUSUMres	(.4)16.571	(.7)**5.706**	(1)**3.251**	(1.2)**2.318**	(1.6)1.855	(1.9)1.564
	CUSUMcla	(.25)16.898	(.5)6.142	(.5)3.538	(.75)2.476	(.75)1.918	(1)1.581
-0.2	EWMAmod	(.05)21.735	(.2)7.782	(.2)4.361	(.4)2.846	(.6)2.120	(.6)1.647
	EWMAres	(.05)**21.711**	(.2)**7.675**	(.3)4.207	(.5)2.820	(.6)2.109	(.8)1.669
	CUSUMmod	(.3)23.410	(.5)7.878	(.7)4.246	(1)2.818	(1.4)2.080	(1.5)**1.645**
	CUSUMres	(.3)22.945	(.6)7.786	(.9)**4.179**	(1.1)**2.770**	(1.4)**2.065**	(1.7)1.646
	CUSUMcla	(.25)23.160	(.5)7.944	(.75)4.314	(.75)2.865	(1)2.105	(1.25)1.664
0.0	EWMAmod	(.05)**28.766**	(.1)**10.333**	(.2)5.501	(.4)3.522	(.5)2.497	(.7)1.865
	EWMAres	(.05)**28.766**	(.1)**10.333**	(.2)5.501	(.4)3.522	(.5)2.497	(.7)1.865
	CUSUMmod	(.3)31.480	(.5)10.519	(.8)**5.458**	(1)**3.413**	(1.3)**2.390**	(1.5)**1.792**
	CUSUMres	(.3)31.480	(.5)10.519	(.8)**5.458**	(1)**3.413**	(1.3)**2.390**	(1.5)**1.792**
	CUSUMcla	(.3)31.480	(.5)10.519	(.8)**5.458**	(1)**3.413**	(1.3)**2.390**	(1.5)**1.792**
0.2	EWMAmod	(.025)38.539	(.1)**13.590**	(.2)7.136	(.4)4.493	(.4)3.073	(.8)2.166
	EWMAres	(.025)**38.520**	(.1)13.677	(.2)7.244	(.3)4.537	(.4)3.104	(.6)2.203
	CUSUMmod	(.3)42.393	(.5)14.353	(.7)7.314	(1)4.409	(1.2)2.925	(1.5)2.068
	CUSUMres	(.2)41.878	(.4)14.325	(.6)7.283	(.8)4.394	(1.1)2.903	(1.3)2.038
	CUSUMcla	(.25)41.473	(.5)14.067	(.75)**7.119**	(1)**4.293**	(1.5)**2.829**	(1.75)**1.988**
0.4	EWMAmod	(.025)**51.486**	(.1)**18.815**	(.2)9.787	(.2)5.977	(.4)3.868	(.8)2.552
	EWMAres	(.025)51.564	(.05)19.123	(.1)10.057	(.2)6.145	(.3)4.116	(.5)2.822
	CUSUMmod	(.3)58.273	(.5)20.300	(.7)10.292	(1)6.040	(1.2)3.807	(1.5)2.516
	CUSUMres	(.2)58.824	(.3)20.120	(.5)10.226	(.6)6.022	(.8)3.819	(1.1)2.503
	CUSUMcla	(.25)56.499	(.5)19.422	(.75)**9.691**	(1.25)**5.642**	(1.5)**3.537**	(2)**2.301**
0.6	EWMAmod	(.025)**74.066**	(.05)**27.286**	(.1)**14.185**	(.2)8.409	(.4)5.238	(1)3.007
	EWMAres	(.01)74.371	(.05)27.800	(.1)14.826	(.1)9.272	(.2)6.057	(.3)4.108
	CUSUMmod	(.2)84.833	(.5)31.238	(.7)16.064	(1)9.242	(1.3)5.468	(1.5)3.303
	CUSUMres	(.1)83.026	(.2)30.347	(.3)15.561	(.4)9.169	(.6)5.708	(.8)3.562
	CUSUMcla	(.25)80.998	(.5)28.636	(.75)14.225	(1.25)**7.985**	(1.75)**4.737**	(2)**2.860**
0.8	EWMAmod	(.01)**119.677**	(.025)**47.398**	(.05)24.982	(.2)14.322	(1)7.494	(1)**3.860**
	EWMAres	(.01)120.784	(.025)48.895	(.05)26.789	(.1)17.027	(.1)11.278	(.2)7.897
	CUSUMmod	(.2)144.146	(.4)60.336	(.7)32.732	(1.3)16.893	(1.4)8.908	(1.5)4.844
	CUSUMres	(.1)149.781	(.1)55.091	(.2)30.236	(.2)17.734	(.3)11.289	(.4)7.189
	CUSUMcla	(.25)133.329	(.5)50.135	(1)**24.878**	(1.5)**13.501**	(2)**7.299**	(2)4.147

7 Summary

The analysis of many industrial datasets has shown that autocorrelation is a widespread pattern of data (Alwan (1989)). If the analyst is unaware of the dependence in the data and he applies standard control charts then more false signals are obtained as supposed and very long delays in detecting the change are frequently observed, respectively. Therefore the classical control charts cannot be applied directly. It is necessary to use the time series structure for constructing new control schemes. Furthermore it has turned out that the detection of structural changes is of great importance in other fields like environmental sciences, finance, medicine, etc. (e.g., Högel (2000), Severin and Schmid (1999)). The extension of the classical charts to time series provides completely new fields of applications and makes these methods more attractive.

In the meantime several control schemes for time series are available. It can be distinguished between two attempts, residual charts and modified control charts. For each type control procedures with memory have been introduced (EWMA, CUSUM). As for independent variables no chart turns out to be overall the best. For positive autocorrelation we recommend (by using the ARL as performance measure) the use of the modified EWMA chart while for negative autocorrelation the classical EWMA chart applied to the residuals or the classical CUSUM chart applied to the residuals must be favored.

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